

Wearable sEMG fatigue monitoring on roofers in hot environments

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1 Abstract

Nepali migrants build much of the Gulf, roofing through summer afternoons on short contracts, with musculoskeletal disorders carried home rather than recorded.

Muscle fatigue precedes those disorders and the falls that injure roofers. Surface EMG (sEMG) tracks fatigue as it builds, before a worker feels it, so a body-worn sensor could warn in time. Nearly all of that evidence comes from cool laboratories.

We ran one protocol twice: indoors on a 30° platform, then at 33–39 °C on an outdoor roof mock-up. Nine of ten channels were usable indoors, four outdoors, and the model fell to 49%, below a 76% majority-class guess. Heat and sweat took the leg channels first, so placement decides whether a study measures fatigue or noise. Nepal builds the same roofs, with no such field data.



Gulf towers



shift change in the heat



roof work at home

2 Methodology

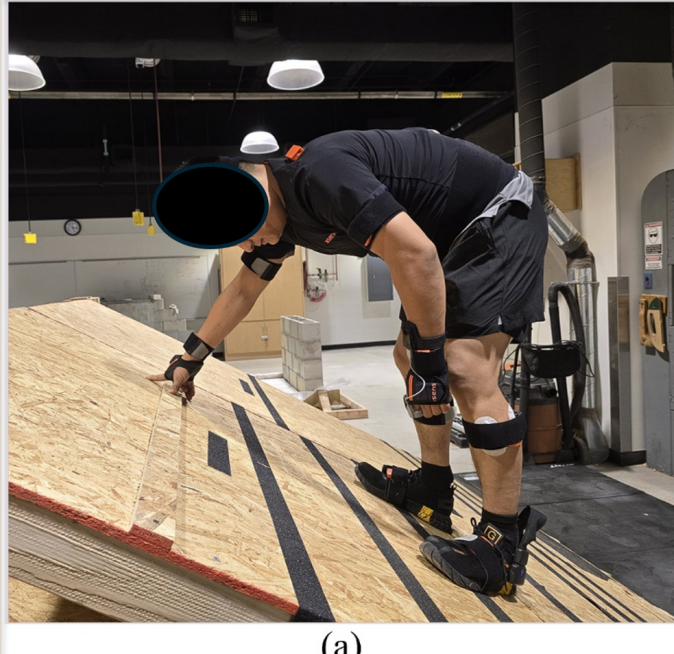
In the laboratory

A channel counts only if it passes all three checks

1 Above the noise floor
bigger than the sensor's own
3–5 μV noise

2 Above its own rest
working signal beats the same
channel at rest

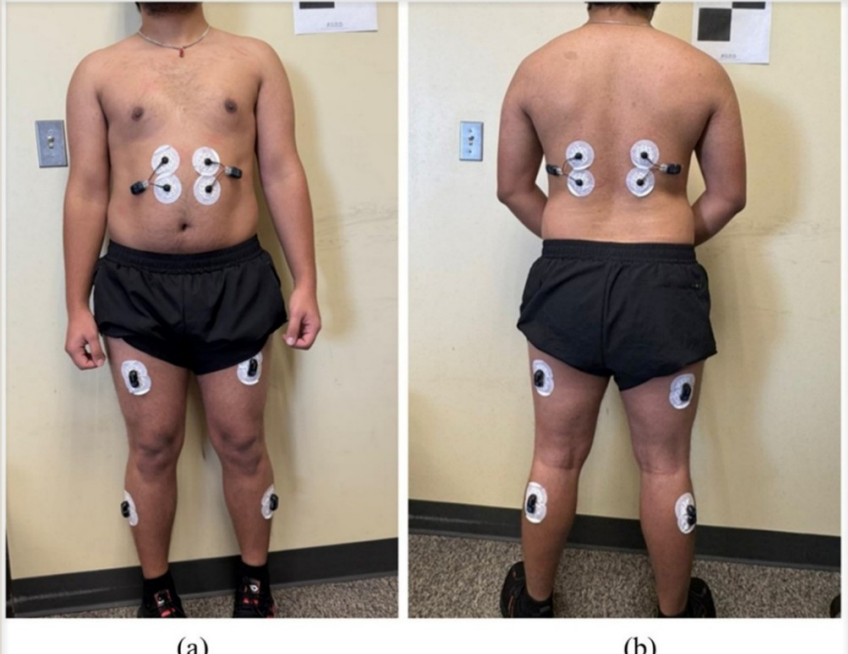
3 In the muscle band
median frequency between 50
and 150 Hz



(a)



(b)



(a)

(b)

the same 30° platform, indoors

ten muscles, trunk to calf

13 laboratory participants · 9 roof workers · Texas State University IRB #9962

On an outdoor roof mock-up in the heat



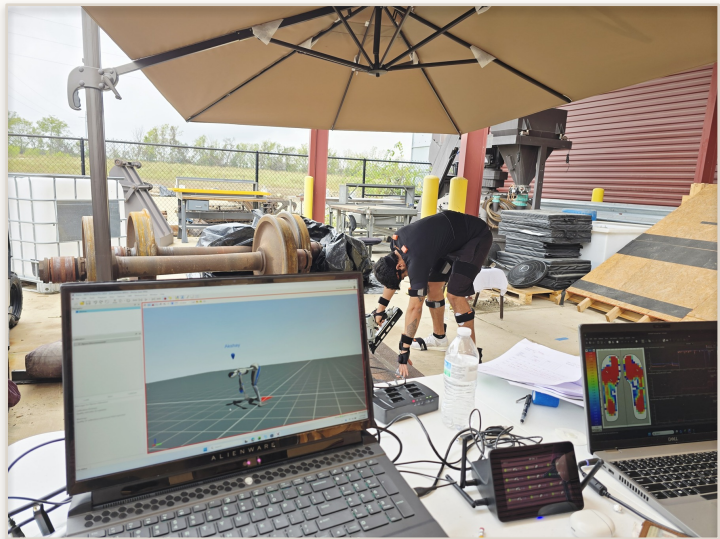
our roof platform



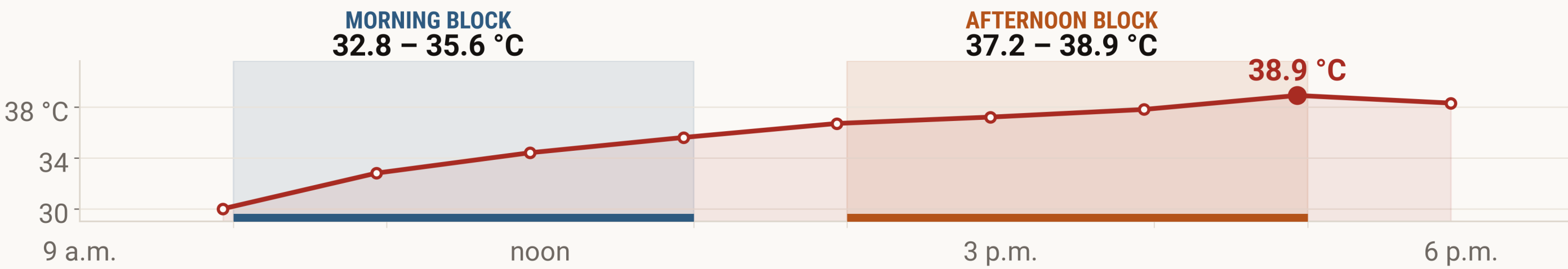
on the worker



the sensor itself



the field station



One session day: 32.8 °C through the morning block, 38.9 °C by five. NOAA station KHYI, San Marcos, 14 August 2025.

3 Results

9→4

LABORATORY OUTDOORS

sensor channels we could use

out of ten

2.3×→1.3×

LABORATORY OUTDOORS

signal working vs resting

higher is easier to read

9→19

LABORATORY OUTDOORS

signal while resting

μV at the lower back

20→8

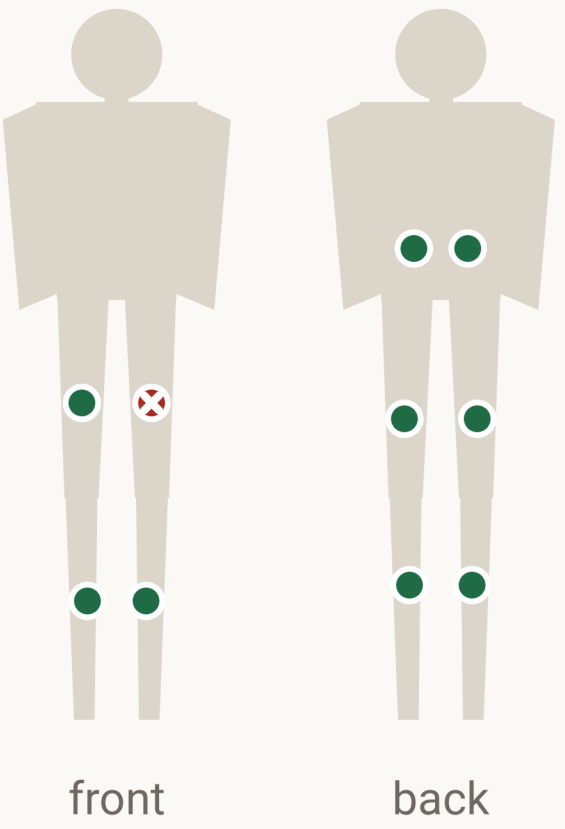
POSSIBLE WE CAN READ

muscle-and-posture tests

kneeling and stooping

LABORATORY

22 °C



front

back

9 of 10 usable

The trunk sensors mostly held.
The leg sensors dropped
into noise.

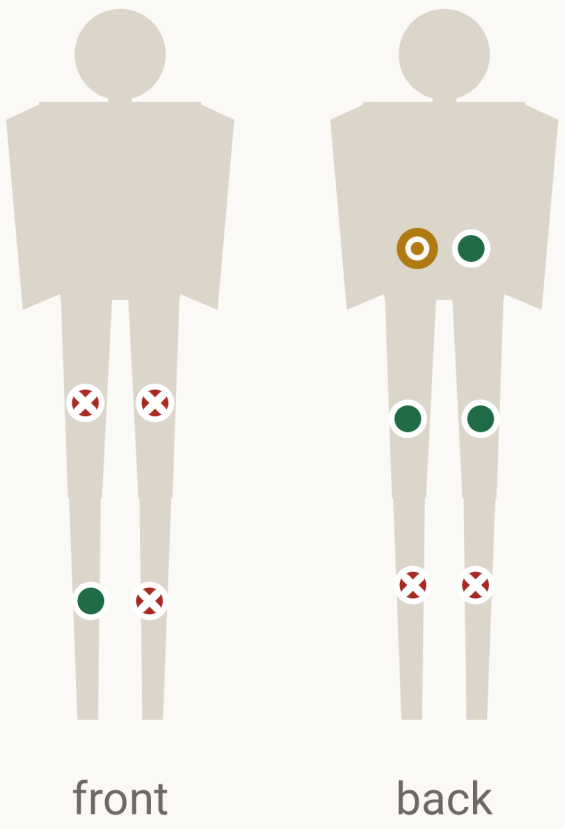
EACH DOT IS ONE SENSOR CHANNEL

- usable
- marginal
- noise only

front and back of the body

OUTDOORS

33–39 °C



front

back

4 of 10 usable



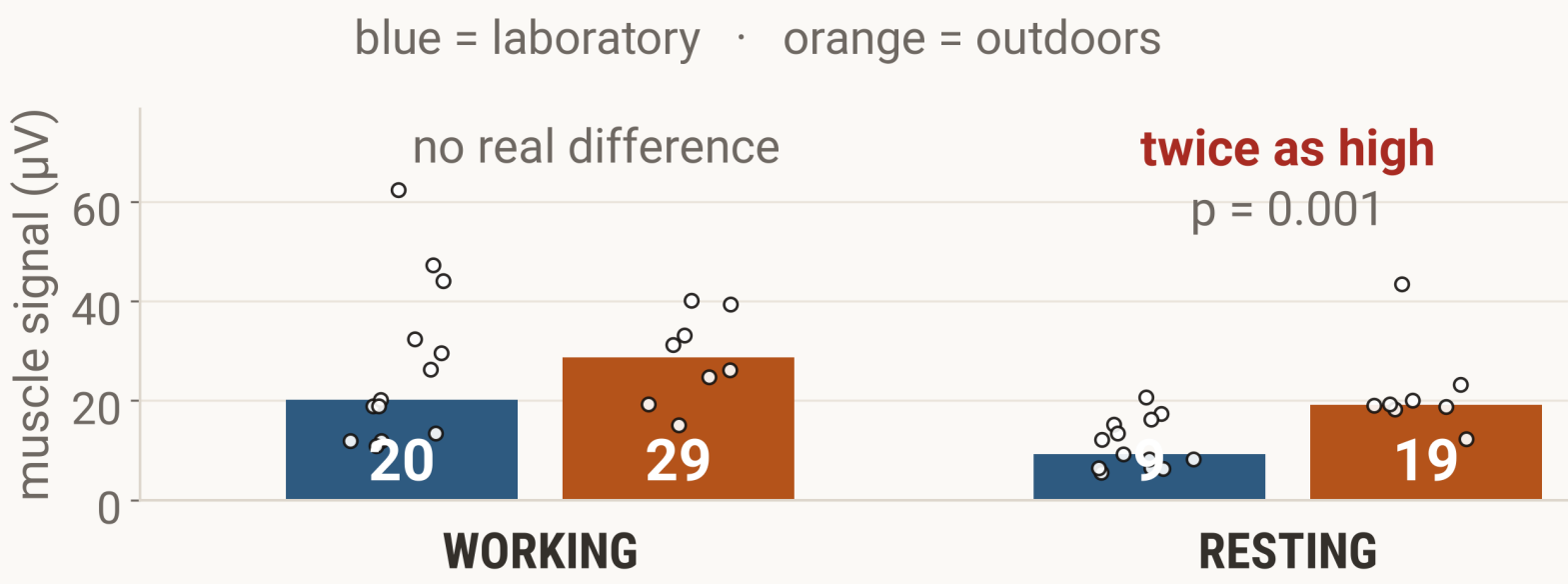
Illustration (AI generated)

Why this happened

Sweat breaks the contact, so the signal gets smaller.

Heat and holding a posture lift the resting level.

Small muscles start closest to the noise, so they cross into it first.



Rest is no longer rest

19 μV on the roof against 9 μV in the lab.

Each dot is one worker.

8 of 20

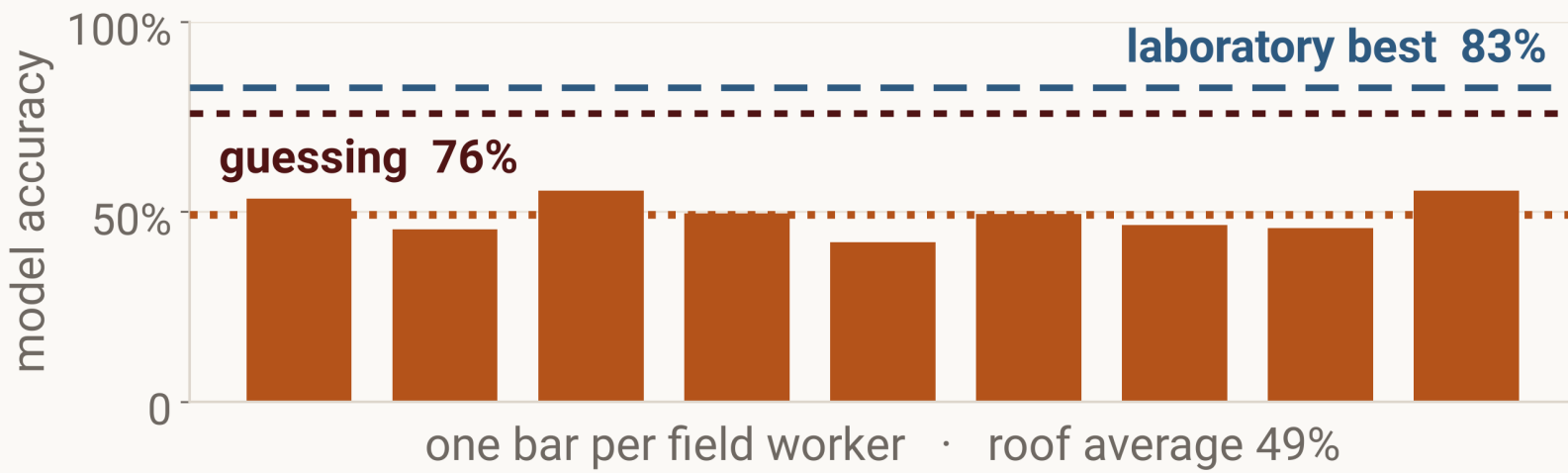
muscle-and-posture tests we can still read



ten sensor channels

Most comparisons are gone

the leg muscles that matter for
balance are the ones we lost



The model inherits the damage

trained one worker at a time, it lands at
49% on roof data

4 Lessons Learned

Trunk first

both sides of the lower back, plus one leg
channel

Check the signal live

a dead channel cannot be fixed later

Teach the sensor check

at vocational centres and pre-departure
briefings

Where Nepal stands: no roof-worn field data exists for Nepali workers. The evidence base is Gulf-employer reports and Western labs, so these lessons are where Nepali researchers can start, on placement and on what a study can claim.

5 Conclusion

A sensor setup that works in a cool laboratory did not survive a hot roof: nine usable channels became four, the leg channels went first, and the fatigue model fell to 49%, below guessing.

Field sEMG studies should be validated where the work happens: place the lower-back channels first, check every channel live before recording, and report which channels survived. In Nepal, where no roof-worn data exist, these rules are a starting point for the first field data set.

22 workers · 30° platform · one processing pipeline · every number here is computed from the recordings at print time · Limitation: different people in each setting, 13 in the laboratory and 9 on the roof, not the same workers tested twice
Reference: Acharya, S., Kisi, K., Gautam, S. R., Mahmud, T., & Kayastha, R. (2025). A high-performance hybrid transformer–LSTM–XGBoost model for sEMG-based fatigue detection in simulated roofing postures. Buildings, 15(17), 3005.

Gulf and Nepal photographs: Imre Solt (CC BY-SA 3.0) and Adam Jones (CC BY-SA 2.0), via Wikimedia Commons.

Sweat-on-the-electrode image: made with Google Flow (Nano Banana 2).